

4 First-Order Scheme

4.3 Numerical experiments

We assess the first-order discretization by a manufactured-solution test on $\Omega = (0, 1)^3$ with $T = 2$. Each Cartesian partition into $K \times K \times K$ cubes is subdivided into tetrahedra, with $K = 4, 8, 16, 32$ and $\Delta t = 1/K = h/\sqrt{3}$, where h is the maximum tetrahedron diameter. The same test is used for the second-order scheme below. Space and time are refined together; this experiment does not isolate the temporal convergence order.

Manufactured solution. Define

$$q(s) = 64s^3(1-s)^3, \quad B(x, y, z) = q(x)q(y)q(z), \quad S(x, y, z) = \sin^2(\pi x) \sin^2(\pi y) \sin^2(\pi z),$$

and let $a = (1, 2, 3)^\top$, $b = (1, -1, 1/2)^\top$, and $c = (1, 2, -1)^\top$. The prescribed fields are

$$\begin{aligned} \mathbf{u} &= \frac{1 + 0.1t}{C_u} \nabla B \times a, & \boldsymbol{\omega} &= \frac{1 - 0.05t}{C_\omega} Bb, & \mathbf{m} &= \frac{1 + 0.05t}{C_m} Sc, \\ \varphi &= \frac{1 - 0.05t}{C_\varphi} B, & \mathbf{H} &= \nabla \varphi, & p &= \frac{1 + 0.05t}{C_p} \prod_{s \in \{x, y, z\}} \cos(2\pi s), \end{aligned}$$

where the fixed normalization constants used in the computations are

$$\begin{aligned} C_u &= 363.275188821, & C_\omega &= 20.6500992827, & C_m &= \frac{\sqrt{81 + 108\pi^2}}{16}, \\ C_\varphi &= 13.7667328551, & C_p &= 0.353553390593. \end{aligned}$$

The auxiliary fields are $\mathbf{k} = \text{curl } \mathbf{m}$, $\mathbf{z} = \mathbf{u} \times \mathbf{m}$, $\vartheta = -\Delta \varphi$, and $\varpi = \mathbf{u} \cdot \mathbf{m}$. We set $\mathbf{H}_e = -\mu_0(\mathbf{m} + \mathbf{H})$ and use the zero-mean modified pressure $\tilde{p} = \mathcal{P}_0[p - \mu_0 \mathbf{m} \cdot \mathbf{H}/2]$, where $\mathcal{P}_0 v = v - |\Omega|^{-1} \int_\Omega v$. All physical parameters $\rho, \kappa, \eta, \zeta, \mu_0, \sigma, \eta', \lambda', \tau, \chi_0$ are set to one.

This is a forced verification problem. The momentum, angular-momentum, and magnetization sources $\mathbf{f}, \mathbf{g}, \mathbf{r}$ are obtained by substituting the prescribed fields into the corresponding continuous equations. Their weak right-hand sides are $(\mathbf{f}, \mathbf{v}_h)$, $(\mathbf{g}, \mathbf{s}_h)$, and $(\mathbf{r}, \mathbf{F}_h)$; the magnetic-potential right-hand side is

$$\left(-\partial_t \mathbf{H}_e - \frac{\mu_0 + \chi_0}{\tau \mu_0} \mathbf{H}_e + \sigma \nabla \text{div } \mathbf{H}_e - \mu_0 \mathbf{r}, \nabla \psi_h \right).$$

The fields satisfy the homogeneous boundary traces used in the experiment: zero velocity and angular velocity, zero normal magnetization, zero tangential $\mathbf{z}, \mathbf{k}, \mathbf{H}$, and zero $\varphi, \vartheta, \varpi$. In particular, q, q', q'' vanish at both endpoints and $S, \nabla S$ vanish on the cube boundary. The new ϑ and ϖ couplings are not identically zero for this test.

Discretization and implementation. The first-order runs use the MINI velocity element, continuous piecewise linear pressure and angular velocity, lowest-order Raviart–Thomas magnetization and first-family Nedelec $\mathbf{z}, \mathbf{k}, \mathbf{H}$, and continuous piecewise linear $\varphi, \vartheta, \varpi$. The initial velocity is obtained by a discrete Stokes projection. After interpolating the initial magnetization into its RT space, the initial potential is determined from the compatible magnetic Poisson problem. The initial $\mathbf{k}, \mathbf{z}, \vartheta, \varpi$ satisfy their discrete mass-projection relations. Pressure is normalized by its finite-element integral mean.

Computations use FEniCSx/DOLFINx 0.10.0 [1] with PETSc and MUMPS. The five-block Algorithm 3 is evaluated through matrix-free Schur-complement actions. The K_1, K_3, K_5 subproblems use MUMPS LU; the S_2, S_4 systems use FGMRES with LU preconditioners. The MUMPS ordering control is ICNTL(7) = 7. The Schur tolerances are 10^{-10} relative and 10^{-12} absolute. All first-order cases and the second-order cases up to $K = 16$ use 48 MPI processes with one thread each; the second-order $K = 32$ case uses 32 MPI processes with four threads each on a 1-TB-class node. Source terms are evaluated at quadrature points, with assembly quadrature degree 12. In the error calculations, the analytic values and derivatives are independently represented in a degree-6 DG space and integrated with degree-12 quadrature; the numerical solution is evaluated in its original finite-element space.

As implementation checks, the two schemes were compared with a monolithic solve of the same matrix at the first two time levels on $K = 4$, modulo the same pressure nullspace; the normalized solution differences were below 1.1×10^{-12} . An independent transcription of the nine weak equations on $K = 2$, using random finite-element fields and both unit and non-unit parameters, also agreed with the production matrices and right-hand sides to scaled differences below 1.1×10^{-16} . These checks concern algebraic and implementation consistency, rather than replacing the convergence test.

Error measures. For a field v and its specified norm X , we report

$$E_X(v) = \frac{\|v - v_h\|_X}{\|v\|_X}, \quad \text{rate}(K \rightarrow 2K) = \log_2 \frac{E_X(v; K)}{E_X(v; 2K)}.$$

The norms are full H^1 for $\mathbf{u}, \boldsymbol{\omega}, \varphi$, L^2 for $\tilde{p}, \mathbf{k}$, $H(\text{div})$ for $\mathbf{m}$, and $H(\text{curl})$ for $\mathbf{H}$. Thus the graph norms combine the squared L^2 and derivative terms before taking the square root. Pressure errors use zero-mean numerical and exact fields. First-order errors are evaluated at T . We also report C_H , the maximum of $|\text{curl } \mathbf{H}_h|$ over the degree-6 DG interpolation points, as a sampled estimate of $\|\text{curl } \mathbf{H}_h\|_{L^\infty}$. It is an absolute diagnostic, not a relative error or a rigorous continuous supremum.

First-order results. Table 1 gives the errors and observed orders; Figure 1 shows the corresponding log-log error curves. On the finest refinement interval, the rates for $\mathbf{u}, \boldsymbol{\omega}, \mathbf{m}, \mathbf{k}, \varphi, \mathbf{H}$ are 1.019, 0.994, 0.999, 0.997, 1.018, and 1.017, respectively. The modified pressure exhibits a rate of 2.065 in this test. The sampled curl diagnostic remains between 1.22×10^{-13} and 8.51×10^{-12} , consistent with the discrete gradient recovery of the magnetic field up to numerical error.

Table 1: Relative errors and observed rates for the first-order scheme, $T = 2$ and $\Delta t = 1/K$. C_H is the sampled absolute curl diagnostic.

(a) Velocity, pressure, angular velocity, and magnetization								
K	$\mathbf{u}: H^1$		$\tilde{p}: L^2$		$\boldsymbol{\omega}: H^1$		$\mathbf{m}: H(\text{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	9.6554×10^{-1}	—	5.6019×10^{-1}	—	6.4182×10^{-1}	—	3.9160×10^{-1}	—
8	4.3070×10^{-1}	1.165	1.0260×10^{-1}	2.449	3.5671×10^{-1}	0.847	1.9873×10^{-1}	0.979
16	1.9858×10^{-1}	1.117	2.1172×10^{-2}	2.277	1.8331×10^{-1}	0.960	9.9738×10^{-2}	0.995
32	9.8002×10^{-2}	1.019	5.0596×10^{-3}	2.065	9.2018×10^{-2}	0.994	4.9919×10^{-2}	0.999
(b) Auxiliary curl field, potential, magnetic field, and curl diagnostic								
K	$\mathbf{k}: L^2$		$\varphi: H^1$		$\mathbf{H}: H(\text{curl})$		C_H	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	5.1511×10^{-1}	—	6.5848×10^{-1}	—	6.6223×10^{-1}	—	1.2220×10^{-13}	—
8	2.6813×10^{-1}	0.942	3.7678×10^{-1}	0.805	3.7965×10^{-1}	0.803	5.1946×10^{-13}	—
16	1.3523×10^{-1}	0.988	1.8757×10^{-1}	1.006	1.8935×10^{-1}	1.004	2.0267×10^{-12}	—
32	6.7750×10^{-2}	0.997	9.2620×10^{-2}	1.018	9.3572×10^{-2}	1.017	8.5067×10^{-12}	—

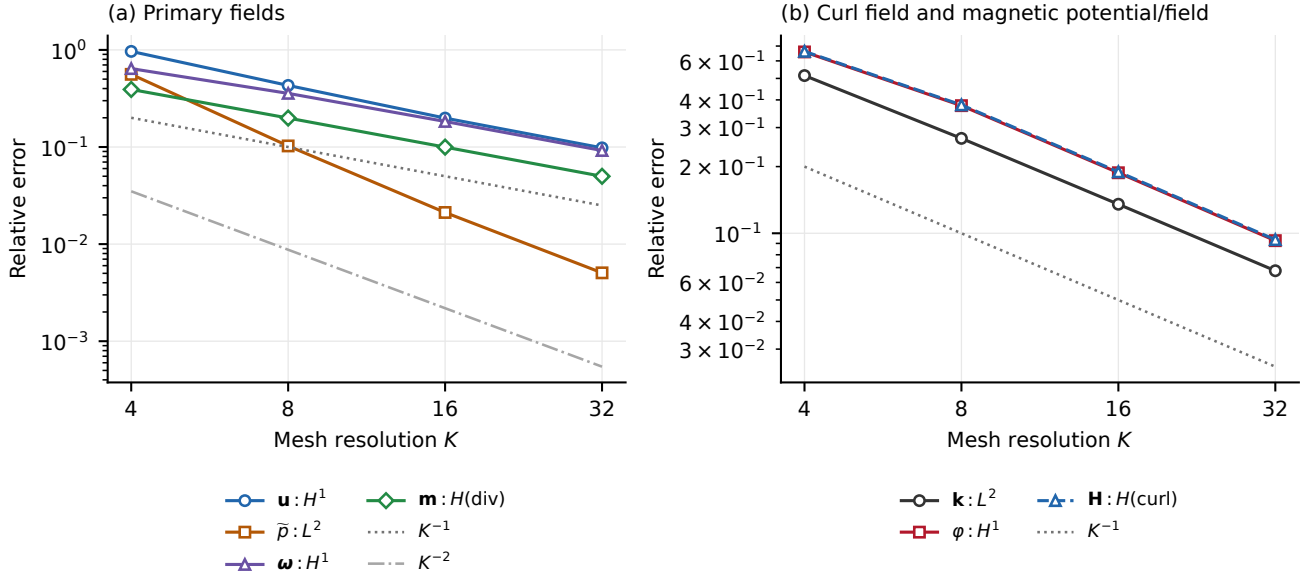


Figure 1: Relative errors for the first-order discretization under joint refinement with $\Delta t = 1/K$. (a) Velocity, modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is the mesh resolution $K = 4, 8, 16, 32$, so refinement proceeds from left to right. Reference lines proportional to K^{-1} and K^{-2} indicate first- and second-order decay with mesh refinement, not independently measured temporal orders.

5 Second-Order Scheme

5.3 Numerical results

We apply the second-order discretization to the same manufactured solution, forcing, parameters, and meshes specified in Section 4.3, retaining $T = 2$ and $\Delta t = 1/K$. Velocity and angular velocity now use continuous piecewise quadratic elements, pressure remains continuous piecewise linear, magnetization uses RT1, and $\mathbf{z}, \mathbf{k}, \mathbf{H}$ use first-family Nedelec elements one degree above the lowest order. The scalar fields $\varphi, \vartheta, \varpi$ are continuous piecewise quadratic. The corresponding RT and Nedelec Basix degree is two.

The first time step is solved by a Broyden quasi-Newton iteration with relative tolerance 10^{-9} and a maximum of 12 iterations. The accepted state is checked by reassembling the nonlinear midpoint equations. Subsequent time steps use the prescribed extrapolated coefficients and require one five-block linear solve per step. The $\mathbf{z}_h$ and ϖ_h slots represent the current midpoint-stage projections; they are not averaged again with the preceding stage.

The primary fields are compared at T . The pressure slot represents the averaged modified pressure, so its reference is

$$\tilde{p}_{\text{ref}}^{N-1/2} = \frac{\tilde{p}(T) + \tilde{p}(T - \Delta t)}{2},$$

with the same zero-mean convention as above. This reference is not identified with either the endpoint pressure or the analytic pressure evaluated directly at the midpoint.

Table 2 and Fig. 2 show that the finest-grid rates for $\mathbf{u}, \tilde{p}, \boldsymbol{\omega}, \mathbf{m}, \varphi, \mathbf{H}$ are 2.166, 2.051, 1.981, 2.007, 1.968, and 1.968, respectively. The L^2 error of $\mathbf{k}$ decreases approximately at first order, with a finest-grid rate of 1.015; second-order convergence is therefore not claimed for every auxiliary quantity. The sampled curl diagnostic ranges from 7.30×10^{-13} to 6.96×10^{-11} . For $K = 32$, all 64 steps reach $T = 2$, and the final full-system relative algebraic residual is 2.08×10^{-13} . These observations support the reported joint-refinement behavior. Because the mesh and time step vary together and the primary exact fields have linear temporal amplitudes, they do not constitute a separate test of pure temporal accuracy or of energy stability.

Table 2: Relative errors and observed rates for the second-order scheme, $T = 2$ and $\Delta t = 1/K$. C_H is the sampled absolute curl diagnostic.

(a) Velocity, pressure, angular velocity, and magnetization								
K	$\mathbf{u}: H^1$		$\tilde{p}: L^2$		$\boldsymbol{\omega}: H^1$		$\mathbf{m}: H(\text{div})$	
	Error	Rate	Error	Rate	Error	Rate	Error	Rate
4	5.6873×10^{-1}	—	4.2000×10^{-1}	—	2.1381×10^{-1}	—	1.1937×10^{-1}	—
8	1.5090×10^{-1}	1.914	9.4815×10^{-2}	2.147	6.2017×10^{-2}	1.786	3.1613×10^{-2}	1.917
16	2.4805×10^{-2}	2.605	2.0872×10^{-2}	2.184	1.6320×10^{-2}	1.926	7.8383×10^{-3}	2.012
32	5.5259×10^{-3}	2.166	5.0382×10^{-3}	2.051	4.1345×10^{-3}	1.981	1.9494×10^{-3}	2.007
(b) Auxiliary curl field, potential, magnetic field, and curl diagnostic								
K	$\mathbf{k}: L^2$		$\varphi: H^1$		$\mathbf{H}: H(\text{curl})$		C_H	
	Error	Rate	Error	Rate	Error	Rate		
4	2.1008×10^{-1}	—	2.0119×10^{-1}	—	2.0303×10^{-1}	—	7.3001×10^{-13}	
8	9.5804×10^{-2}	1.133	6.0098×10^{-2}	1.743	6.0718×10^{-2}	1.742	3.9064×10^{-12}	
16	4.6084×10^{-2}	1.056	1.6119×10^{-2}	1.899	1.6289×10^{-2}	1.898	1.8492×10^{-11}	
32	2.2804×10^{-2}	1.015	4.1189×10^{-3}	1.968	4.1627×10^{-3}	1.968	6.9611×10^{-11}	

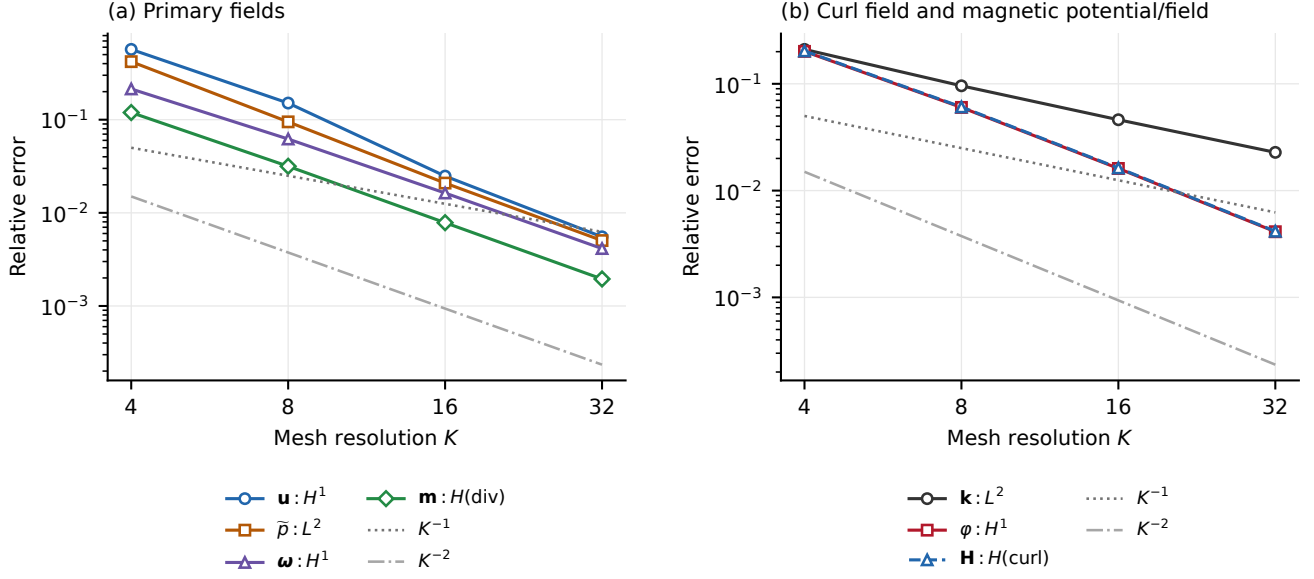


Figure 2: Relative errors for the second-order discretization using the same manufactured solution and $\Delta t = 1/K$. (a) Velocity, averaged modified pressure, angular velocity, and magnetization. (b) Auxiliary curl field, magnetic potential, and magnetic field. The horizontal coordinate is $K = 4, 8, 16, 32$, increasing from left to right. Reference lines proportional to K^{-1} and K^{-2} distinguish the nearly quadratic decay of the principal errors from the approximately first-order L^2 decay for $\mathbf{k}$.

References

- [1] I. A. Baratta, J. P. Dean, J. S. Dokken, M. Habera, J. S. Hale, C. N. Richardson, M. E. Rognes, M. W. Scroggs, N. Sime, and G. N. Wells, DOLFINx: The next generation FEniCS problem solving environment, 2023.